Inference on Policy Value under Admissibility-Constrained Optimal Treatment Rules*

Zhenxiao Chen†

July 13, 2026

Abstract

Contents

*Thank everyone.

†Department of Economics, University of Pennsylvania, USA, zxchen@upenn.edu.

1 Introduction

1.1 Questions

How to handle smoother?

1.2 Why Point-Wise Admissibility?

In many economic problems, a treatment will usually lead to multiple consequences, some of them are favorable to the social planner, while others are not.

Among those unfavorable consequences, some of them could be compensated by a good outcome, while others are hard to trade off. Consider the example of the long-term and short-term outcomes. The social planner may be able to trade off some short term loss for long-term benefit for workers in a job training program. Now consider the other scenario where the treatment is a drug and two outcomes are health benefit and side effects, then a social planner might not want to a bad side effect in order to obtain a good health outcome.

The nature of non-compensatory and compensatory outcomes due to a treatment leads to different types social planner's problem, one with a soft constraint, while the other with a hard threshold constraint

Non-compensatory makes it

global to local

And a hard threshold is what leads to point-wise admissibility.

The policy class is constrained to be decision trees. This is studied in Atehy's paper. And there is Christensen's paper on generated regressors for latent covariates using machine learning. So splitting can happen based on latent covaraites. If the

necessary condtion

in the ethics of modern medicine, the Hippocratic principle of "do no harm" remains influential

If we think of the policy learning problem as eliminating those people who will not benefit as much

1.3 Motivation 1: Psychology

1.4 Motivation 3: Marketing

1.5 Motivation 2: Micro theory

There is a literature on sequential screening. The idea is that a social planner has an ordered-checklist of finite length regarding proprieties of the target population, and she goes through the checklist sequentially to discriminate among available alternatives.

One example is a decision maker who considers a treatment allocation task successful only if treated unit not only benefit from the intervention directly but also the benefit will be long lasting. In other words, a positive long-term conditional average treatment effect is a necessary condition for a successful treatment allocation.

Usually, the order of elimination matters, but under the condition that the checklist has a finite length and the survivor set is nonempty after each round of elimination, the agent faces a utility function and preference.

What hides underneath the premise of sequential elimination is the social planner's tendency to not trade off between earlier and later properties.

1.6 Differential Geometry Preliminaries

2 Examples

2.1 Population Summary Statistics

2.2 Severe Side Effects

In drug assignment, efficacy and side effects need not enter the planner's problem symmetrically. Mild side effects may be treated as ordinary costs and incorporated into a net health-benefit margin. Severe adverse events, however, often operate as safety constraints rather than as prices in a welfare index. A regulator or public health planner may be unwilling to recommend a drug to a subgroup whose risk of a serious side effect exceeds an acceptable level, even if the expected therapeutic gain is large. This reflects a minimum safety requirement: treatment must be beneficial, but it must also be safe enough.

2.3 Strict Overlap

In a population where only overlap is met, it might be sensible to focus on a subpopulation where strict overlap is satisfied. This is because the asymptotic variance is controlled.

2.4 Dynamic Treatment Regime (Chen, Austern and Syrgkanis, 2023)

We consider a two-period treatment assignment problem. Let S denote the first-period state, let $T_1 \in \{0,1\}$ denote the first-period treatment, let X denote the second-period state, let $T_2 \in \{0,1\}$ denote the second-period treatment, and let Y denote the final outcome. A first-period policy is a measurable map $\pi_1 : \mathcal{S} \to \{0,1\}$, and a second-period policy is a measurable map $\pi_2 : \mathcal{X} \to \{0,1\}$. For static treatment assignments $(\tau_1, \tau_2) \in \{0,1\}^2$, write $Y^{(\tau_1,\tau_2)}$ for the potential final outcome. We also write $X^{(\tau_1)}$ for the potential second-period state that would be realized under first-period treatment τ_1.

Given a second-period policy π_2, define the potential outcome under first-period action τ_1 and second-period policy π_2 by

$$Y^{(\tau_1,\pi_2)} := \sum_{\tau_2 \in \{0,1\}} \mathbb{1}\left\{\pi_2\left(X^{(\tau_1)}\right) = \tau_2\right\} Y^{(\tau_1,\tau_2)}.$$

Similarly, for a pair of policies $\pi = (\pi_1, \pi_2)$, define

$$Y^{(\pi_1,\pi_2)} := \sum_{\tau_1 \in \{0,1\}} \mathbb{1}\{\pi_1(S) = \tau_1\} Y^{(\tau_1,\pi_2)}.$$

The policy value is

$$V(\pi_1, \pi_2) := \mathbb{E}\left[Y^{(\pi_1,\pi_2)}\right].$$

The object of interest is an optimal regime

$$(\pi_1^*, \pi_2^*) \in \arg\max_{(\pi_1,\pi_2)} V(\pi_1, \pi_2),$$

and the corresponding optimal value is denoted by

$$V^* := V(\pi_1^*, \pi_2^*).$$

Assumption 2.1 (Sequential Independence, Markov Sufficiency, and Consistency)**.** *The following conditions hold.*

1. ***Consistency.*** *If* $T_1 = \tau_1$, *then*
$$X = X^{(\tau_1)}.$$
If $(T_1, T_2) = (\tau_1, \tau_2)$, *then*
$$Y = Y^{(\tau_1,\tau_2)}.$$

2. ***First-period sequential independence.***
$$\big(X^{(0)}, X^{(1)}, Y^{(0,0)}, Y^{(0,1)}, Y^{(1,0)}, Y^{(1,1)}\big) \perp T_1 \mid S.$$

3. ***Second-period sequential independence.*** $\tau_1, \tau_2 \in \{0, 1\}$,
$$(Y^{(T_1,0)}, Y^{(T_1,1)}) \perp T_2 \mid X.$$

4. ***Positivity.*** *For all relevant states* s, x *and all* $\tau_1, \tau_2 \in \{0, 1\}$,
$$\mathbb{P}(T_1 = \tau_1 \mid S = s) > 0, \qquad \mathbb{P}(T_2 = \tau_2 \mid S = s, T_1 = \tau_1, X = x) > 0.$$

We use the following notation. Let $d_x := \dim(\mathcal{X})$ and $d_s := \dim(\mathcal{S})$. For $a \in \{0, 1\}$, define
$$p_a(x \mid s) := f_{X|S,T_1=a}(x \mid s), \qquad r(x \mid s) := p_1(x \mid s) - p_0(x \mid s), \qquad m(s) := f_S(s).$$
Let
$$\mu_a(x) := \mathbb{E}[Y \mid X = x, T_2 = a], \qquad \delta(x) := \mu_1(x) - \mu_0(x).$$
Define the second-stage treated region, untreated region, and margin by
$$D_2^+ := \{x \in \mathcal{X} : \delta(x) \geq 0\}, \qquad D_2^- := \{x \in \mathcal{X} : \delta(x) < 0\}, \qquad M_2 := \{x \in \mathcal{X} : \delta(x) = 0\}.$$
The second-stage value function is
$$V_2(x) := \max\{\mu_0(x), \mu_1(x)\} = \mu_0(x) + [\delta(x)]_+.$$
For $a \in \{0, 1\}$, define the continuation value
$$A_a(s) := \mathbb{E}[V_2(X) \mid S = s, T_1 = a] = \int_{\mathcal{X}} V_2(x) p_a(x \mid s)\, \mathrm{d}x.$$

The first-stage contrast is

$$\begin{aligned}\kappa(s) &:= A_1(s) - A_0(s) \\ &= \mathbb{E}[V_2(X) \mid S = s, T_1 = 1] - \mathbb{E}[V_2(X) \mid S = s, T_1 = 0] \\ &= \int_{\mathcal{X}} V_2(x)\{p_1(x \mid s) - p_0(x \mid s)\}\,\mathrm{d}x \\ &= \int_{\mathcal{X}} V_2(x) r(x \mid s)\,\mathrm{d}x.\end{aligned}$$

Define the first-stage treated region, untreated region, and margin by

$$D_1^+ := \{s \in \mathcal{S} : \kappa(s) \geq 0\}, \qquad D_1^- := \{s \in \mathcal{S} : \kappa(s) < 0\}, \qquad M_1 := \{s \in \mathcal{S} : \kappa(s) = 0\}.$$

The strict inequalities in D_1^- and D_2^- encode the tie-breaking convention that action 1 is chosen on the margin.

Proposition 2.2 (Characterization of the Optimal Sequential Regime)**.** *Suppose Assumption 2.1 holds. An optimal second-period policy satisfies*

$$\pi_2^*(x) \in \arg\max_{\tau_2 \in \{0,1\}} \mu_{\tau_2}(x).$$

Moreover, an optimal first-period policy satisfies

$$\pi_1^*(s) \in \arg\max_{\tau_1 \in \{0,1\}} \mathbb{E}[V_2(X) \mid S = s, T_1 = \tau_1].$$

Equivalently, under the tie-breaking convention that action 1 *is chosen whenever the two actions have equal value,*

$$\mathbb{1}\{\pi_2^*(X) = 1\} = \mathbb{1}\{\delta(X) \geq 0\},$$

and

$$\mathbb{1}\{\pi_1^*(S) = 1\} = \mathbb{1}\{\kappa(S) \geq 0\}.$$

The optimal value admits the following decomposition into treatment-path components:

$$\begin{aligned}V^* = \mathbb{E}\big[&Y^{(0,0)}\mathbb{1}\{\pi_1^*(S) = 0, \pi_2^*(X^{(0)}) = 0\}\big] \\ &+ \mathbb{E}\big[Y^{(1,0)}\mathbb{1}\{\pi_1^*(S) = 1, \pi_2^*(X^{(1)}) = 0\}\big] \\ &+ \mathbb{E}\big[Y^{(0,1)}\mathbb{1}\{\pi_1^*(S) = 0, \pi_2^*(X^{(0)}) = 1\}\big] \\ &+ \mathbb{E}\big[Y^{(1,1)}\mathbb{1}\{\pi_1^*(S) = 1, \pi_2^*(X^{(1)}) = 1\}\big].\end{aligned}$$

The total welfare V^* is the value studied in optimal dynamic treatment regime problems such as Chen *et al.* (2023). The individual treatment-path components can also be economically relevant, because they measure the welfare generated by different subpopulations under the optimal regime. The distinction between total welfare and its path components is important for inference: as shown below, the path components generally contain lower-dimensional boundary terms, while these boundary terms cancel in the derivative of the total welfare.

We first analyze the $(1,1)$ component. Define

$$V_{11}^* := \mathbb{E}\big[Y^{(1,1)}\mathbb{1}\{\pi_1^*(S) = 1, \pi_2^*(X^{(1)}) = 1\}\big]\,,$$

and

$$G_{11}(s) := \int_{D_2^+} \mu_1(x) p_1(x \mid s)\,\mathrm{d}x.$$

Then

$$V_{11}^* = \int_{D_1^+} G_{11}(s) m(s)\,\mathrm{d}s = \int_{D_1^+}\int_{D_2^+} \mu_1(x) p_1(x \mid s) m(s)\,\mathrm{d}x\,\mathrm{d}s.$$

The identifying representation of V_{11}^* is obtained as follows:

$$\begin{aligned}
V_{11}^* &= \mathbb{E}\big[Y^{(1,1)}\mathbb{1}\{\pi_1^*(S) = 1, \pi_2^*(X^{(1)}) = 1\}\big] \\
&= \int_{\mathcal{S}}\int_{\mathcal{X}} \mathbb{1}\{\pi_1^*(s) = 1\}\mathbb{1}\{\pi_2^*(x) = 1\}\mathbb{E}[Y^{(1,1)} \mid S = s, X^{(1)} = x] \\
&\qquad\qquad\qquad \times f_{X^{(1)}|S}(x \mid s) m(s)\,\mathrm{d}x\,\mathrm{d}s \\
&= \int_{\mathcal{S}}\int_{\mathcal{X}} \mathbb{1}\{\pi_1^*(s) = 1\}\mathbb{1}\{\pi_2^*(x) = 1\}\mathbb{E}[Y \mid S = s, T_1 = 1, X = x, T_2 = 1] \\
&\qquad\qquad\qquad \times p_1(x \mid s) m(s)\,\mathrm{d}x\,\mathrm{d}s \\
&= \int_{\mathcal{S}}\int_{\mathcal{X}} \mathbb{1}\{\pi_1^*(s) = 1\}\mathbb{1}\{\pi_2^*(x) = 1\}\mathbb{E}[Y \mid X = x, T_2 = 1] \\
&\qquad\qquad\qquad \times p_1(x \mid s) m(s)\,\mathrm{d}x\,\mathrm{d}s \\
&= \int_{\mathcal{S}}\int_{\mathcal{X}} \mathbb{1}\{\delta(x) \geq 0\}\mathbb{1}\{\kappa(s) \geq 0\}\mu_1(x) p_1(x \mid s) m(s)\,\mathrm{d}x\,\mathrm{d}s \\
&= \int_{D_1^+}\int_{D_2^+} \mu_1(x) p_1(x \mid s) m(s)\,\mathrm{d}x\,\mathrm{d}s.
\end{aligned}$$

The second line writes the potential outcome expectation by conditioning on the potential second-period state. The third line uses consistency, sequential independence, positivity, and the identified law $f_{X^{(1)}|S}(x \mid s) = p_1(x \mid s)$. The fourth line uses Markov sufficiency. The last two lines substitute the threshold characterizations of π_1^* and π_2^* and then rewrite the indicator functions as integration regions.

We now compute the pathwise derivative of V_{11}^* with respect to μ_0 and μ_1, holding p_a and m fixed. Let

$$\mu_{a,t} := \mu_a + t\dot{\mu}_a, \qquad a \in \{0, 1\},$$

and define

$$\delta_t(x) := \mu_{1,t}(x) - \mu_{0,t}(x), \qquad \dot{\delta}(x) := \dot{\mu}_1(x) - \dot{\mu}_0(x).$$

Let

$$D_{2,t}^+ := \{x \in \mathcal{X} : \delta_t(x) \geq 0\}, \qquad D_{2,t}^- := \{x \in \mathcal{X} : \delta_t(x) < 0\},$$

and

$$V_{2,t}(x) := \max\{\mu_{0,t}(x), \mu_{1,t}(x)\} = \mu_{0,t}(x) + [\delta_t(x)]_+.$$

For the $(1, 1)$ component, set

$$G_{11,t}(s) := \int_{D_{2,t}^+} \mu_{1,t}(x) p_1(x \mid s)\, \mathrm{d}x.$$

We maintain the following regular-margin conditions for the derivative calculations. The functions δ and κ are continuously differentiable in neighborhoods of M_2 and M_1, respectively; 0 is a regular value of both functions; and

$$\|\nabla_x \delta(x)\| > 0 \quad \text{for all } x \in M_2, \qquad \|\nabla_s \kappa(s)\| > 0 \quad \text{for all } s \in M_1.$$

Thus M_2 and M_1 are smooth $(d_x - 1)$- and $(d_s - 1)$-dimensional level sets, respectively. Under these conditions, the moving-boundary formula gives, for a generic smooth perturbation $h_t = h + t\dot{h}$ and integrand q_t,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\{z : h_t(z) \geq 0\}} q_t(z)\, \mathrm{d}z \bigg|_{t=0} = \int_{\{z : h(z) \geq 0\}} \dot{q}(z)\, \mathrm{d}z + \int_{\{z : h(z) = 0\}} \frac{\dot{h}(z) q(z)}{\|\nabla h(z)\|}\, \mathrm{d}\mathcal{H}^{d-1}(z),$$

with the sign of the boundary term reversed for the region $\{z : h_t(z) < 0\}$.

Applying this formula to $G_{11,t}$ yields

$$\begin{aligned}
\dot{G}_{11}(s) &:= \frac{\mathrm{d}}{\mathrm{d}t} G_{11,t}(s) \bigg|_{t=0} \\
&= \int_{D_2^+} \dot{\mu}_1(x) p_1(x \mid s)\, \mathrm{d}x + \int_{M_2} \frac{\dot{\delta}(x) \mu_1(x) p_1(x \mid s)}{\|\nabla_x \delta(x)\|}\, \mathrm{d}\mathcal{H}^{d_x - 1}(x).
\end{aligned}$$

The first term differentiates the integrand with the region held fixed. The second term is the first-order contribution from the movement of the second-stage decision boundary M_2. This is the step that uses the regular-margin condition for δ; the rest is ordinary differentiation under the integral sign.

Next define

$$\kappa_t(s) := \int_{\mathcal{X}} V_{2,t}(x) r(x \mid s) \, \mathrm{d}x.$$

Using $V_{2,t}(x) = \mu_{0,t}(x) + [\delta_t(x)]_+$, we have

$$\kappa_t(s) = \int_{\mathcal{X}} \mu_{0,t}(x) r(x \mid s) \, \mathrm{d}x + \int_{D_{2,t}^+} \delta_t(x) r(x \mid s) \, \mathrm{d}x.$$

Therefore,

$$\begin{aligned}
\dot{\kappa}(s) &:= \frac{\mathrm{d}}{\mathrm{d}t} \kappa_t(s) \Big|_{t=0} \\
&= \int_{\mathcal{X}} \dot{\mu}_0(x) r(x \mid s) \, \mathrm{d}x + \int_{D_2^+} \dot{\delta}(x) r(x \mid s) \, \mathrm{d}x \\
&\quad + \int_{M_2} \frac{\dot{\delta}(x) \delta(x) r(x \mid s)}{\|\nabla_x \delta(x)\|} \, \mathrm{d}\mathcal{H}^{d_x - 1}(x) \\
&= \int_{\mathcal{X}} \dot{\mu}_0(x) r(x \mid s) \, \mathrm{d}x + \int_{D_2^+} \{\dot{\mu}_1(x) - \dot{\mu}_0(x)\} r(x \mid s) \, \mathrm{d}x \\
&= \int_{D_2^+} \dot{\mu}_1(x) r(x \mid s) \, \mathrm{d}x + \int_{D_2^-} \dot{\mu}_0(x) r(x \mid s) \, \mathrm{d}x.
\end{aligned}$$

The third line contains the moving-boundary term from $D_{2,t}^+$, but it vanishes because $\delta(x) = 0$ on M_2. The fourth line substitutes $\dot{\delta} = \dot{\mu}_1 - \dot{\mu}_0$. The last line is an algebraic regrouping, using that M_2 has zero d_x-dimensional Lebesgue measure under the regular-margin condition.

Finally, define

$$D_{1,t}^+ := \{s \in \mathcal{S} : \kappa_t(s) \geq 0\}, \qquad V_{11,t}^* := \int_{D_{1,t}^+} G_{11,t}(s) m(s) \, \mathrm{d}s.$$

Then

$$
\begin{aligned}
\dot V_{11}^* &:= \frac{\mathrm{d}}{\mathrm{d}t} V_{11,t}^* \bigg|_{t=0} \\
&= \int_{D_1^+} \dot G_{11}(s) m(s)\,\mathrm{d}s + \int_{M_1} \frac{\dot\kappa(s) G_{11}(s) m(s)}{\|\nabla_s \kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s) \\
&= \int_{D_1^+} \left[\int_{D_2^+} \dot\mu_1(x) p_1(x \mid s)\,\mathrm{d}x + \int_{M_2} \frac{\dot\delta(x)\mu_1(x)p_1(x \mid s)}{\|\nabla_x \delta(x)\|}\,\mathrm{d}\mathcal{H}^{d_x-1}(x) \right] m(s)\,\mathrm{d}s \\
&\quad + \int_{M_1} \frac{G_{11}(s)m(s)}{\|\nabla_s \kappa(s)\|} \left[\int_{\mathcal{X}} \dot\mu_0(x) r(x \mid s)\,\mathrm{d}x + \int_{D_2^+} \{\dot\mu_1(x) - \dot\mu_0(x)\} r(x \mid s)\,\mathrm{d}x \right] \mathrm{d}\mathcal{H}^{d_s-1}(s) \\
&= \int_{D_1^+} \int_{D_2^+} \dot\mu_1(x) p_1(x \mid s) m(s)\,\mathrm{d}x\,\mathrm{d}s \\
&\quad + \int_{D_1^+} \int_{M_2} \frac{\dot\delta(x)\mu_1(x)p_1(x \mid s)}{\|\nabla_x \delta(x)\|} m(s)\,\mathrm{d}\mathcal{H}^{d_x-1}(x)\,\mathrm{d}s \\
&\quad + \int_{M_1} \frac{\dot\kappa(s) G_{11}(s) m(s)}{\|\nabla_s \kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s).
\end{aligned}
$$

The first equality uses the moving-boundary derivative for the first-stage region $D_{1,t}^+$ and therefore uses the regular-margin condition for κ. The next equality substitutes the formulas for $\dot G_{11}(s)$ and $\dot\kappa(s)$. The final equality only separates the full-dimensional Lebesgue integral from the two lower-dimensional Hausdorff integrals.

The derivative of V_{11}^* therefore contains three terms. The first is an ordinary interior term: it perturbs μ_1 for states that remain on the $(1,1)$ path. The second is a second-stage boundary term: it accounts for the infinitesimal mass of second-stage states crossing M_2. The third is a first-stage boundary term: it accounts for the infinitesimal mass of first-stage states crossing M_1. The factors $\dot\delta(x)/\|\nabla_x \delta(x)\|$ and $\dot\kappa(s)/\|\nabla_s \kappa(s)\|$ are the normal velocities of the corresponding decision boundaries.

We now show why the total welfare does not contain these lower-dimensional terms, even though its individual path components do. For each $a \in \{0,1\}$, write

$$
\begin{aligned}
A_a(s) &= \mathbb{E}[V_2(X) \mid S = s, T_1 = a] \\
&= \int_{\mathcal{X}} \left[\mu_1(x) \mathbb{1}\{\delta(x) \geq 0\} + \mu_0(x) \mathbb{1}\{\delta(x) < 0\} \right] p_a(x \mid s)\,\mathrm{d}x \\
&= \int_{D_2^+} \mu_1(x) p_a(x \mid s)\,\mathrm{d}x + \int_{D_2^-} \mu_0(x) p_a(x \mid s)\,\mathrm{d}x.
\end{aligned}
$$

Let

$$A_{a,t}(s) := \int_{D_{2,t}^{+}} \mu_{1,t}(x) p_a(x \mid s)\,\mathrm{d}x + \int_{D_{2,t}^{-}} \mu_{0,t}(x) p_a(x \mid s)\,\mathrm{d}x.$$

Applying the moving-boundary formula to both $D_{2,t}^{+}$ and $D_{2,t}^{-}$ gives

$$\begin{aligned}
\dot{A}_a(s) &:= \frac{\mathrm{d}}{\mathrm{d}t} A_{a,t}(s)\bigg|_{t=0} \\
&= \int_{D_2^{+}} \dot{\mu}_1(x) p_a(x \mid s)\,\mathrm{d}x + \int_{M_2} \frac{\dot{\delta}(x)\mu_1(x) p_a(x \mid s)}{\|\nabla_x \delta(x)\|}\,\mathrm{d}\mathcal{H}^{d_x-1}(x) \\
&\quad + \int_{D_2^{-}} \dot{\mu}_0(x) p_a(x \mid s)\,\mathrm{d}x - \int_{M_2} \frac{\dot{\delta}(x)\mu_0(x) p_a(x \mid s)}{\|\nabla_x \delta(x)\|}\,\mathrm{d}\mathcal{H}^{d_x-1}(x) \\
&= \int_{D_2^{+}} \dot{\mu}_1(x) p_a(x \mid s)\,\mathrm{d}x + \int_{D_2^{-}} \dot{\mu}_0(x) p_a(x \mid s)\,\mathrm{d}x \\
&\quad + \int_{M_2} \frac{\dot{\delta}(x)\{\mu_1(x) - \mu_0(x)\} p_a(x \mid s)}{\|\nabla_x \delta(x)\|}\,\mathrm{d}\mathcal{H}^{d_x-1}(x) \\
&= \int_{D_2^{+}} \dot{\mu}_1(x) p_a(x \mid s)\,\mathrm{d}x + \int_{D_2^{-}} \dot{\mu}_0(x) p_a(x \mid s)\,\mathrm{d}x.
\end{aligned}$$

The two second-stage boundary terms enter with opposite signs because $D_{2,t}^{+}$ expands when $D_{2,t}^{-}$ contracts. They cancel because their combined boundary weight is $\mu_1(x) - \mu_0(x) = \delta(x)$, which is zero on M_2. Thus the continuation value $A_a(s)$ has no first-order second-stage boundary contribution.

Since

$$\kappa_t(s) = A_{1,t}(s) - A_{0,t}(s),$$

we also obtain

$$\begin{aligned}
\dot{\kappa}(s) &= \dot{A}_1(s) - \dot{A}_0(s) \\
&= \int_{D_2^{+}} \dot{\mu}_1(x)\{p_1(x \mid s) - p_0(x \mid s)\}\,\mathrm{d}x + \int_{D_2^{-}} \dot{\mu}_0(x)\{p_1(x \mid s) - p_0(x \mid s)\}\,\mathrm{d}x \\
&= \int_{D_2^{+}} \dot{\mu}_1(x) r(x \mid s)\,\mathrm{d}x + \int_{D_2^{-}} \dot{\mu}_0(x) r(x \mid s)\,\mathrm{d}x.
\end{aligned}$$

This is the same expression obtained above from differentiating $\kappa_t(s) = \int V_{2,t}(x) r(x \mid s)\,\mathrm{d}x$. The display is included to make clear that the absence of the M_2 term in $\dot{\kappa}(s)$ is a cancellation result, not a notational omission.

The total optimal welfare is

$$V^* = \int_{D_1^+} A_1(s)m(s)\,\mathrm{d}s + \int_{D_1^-} A_0(s)m(s)\,\mathrm{d}s.$$

For the perturbed total welfare,

$$V_t^* = \int_{D_{1,t}^+} A_{1,t}(s)m(s)\,\mathrm{d}s + \int_{D_{1,t}^-} A_{0,t}(s)m(s)\,\mathrm{d}s.$$

Therefore,

$$\begin{aligned}
\dot{V}^* &:= \frac{\mathrm{d}}{\mathrm{d}t} V_t^* \Big|_{t=0} \\
&= \int_{D_1^+} \dot{A}_1(s)m(s)\,\mathrm{d}s + \int_{M_1} \frac{\dot{\kappa}(s)A_1(s)m(s)}{\|\nabla_s \kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s) \\
&\quad + \int_{D_1^-} \dot{A}_0(s)m(s)\,\mathrm{d}s - \int_{M_1} \frac{\dot{\kappa}(s)A_0(s)m(s)}{\|\nabla_s \kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s) \\
&= \int_{D_1^+} \dot{A}_1(s)m(s)\,\mathrm{d}s + \int_{D_1^-} \dot{A}_0(s)m(s)\,\mathrm{d}s \\
&\quad + \int_{M_1} \frac{\dot{\kappa}(s)\{A_1(s) - A_0(s)\}m(s)}{\|\nabla_s \kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s) \\
&= \int_{D_1^+} \dot{A}_1(s)m(s)\,\mathrm{d}s + \int_{D_1^-} \dot{A}_0(s)m(s)\,\mathrm{d}s \\
&\quad + \int_{M_1} \frac{\dot{\kappa}(s)\kappa(s)m(s)}{\|\nabla_s \kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s) \\
&= \int_{D_1^+} \dot{A}_1(s)m(s)\,\mathrm{d}s + \int_{D_1^-} \dot{A}_0(s)m(s)\,\mathrm{d}s.
\end{aligned}$$

The first two lines apply the moving-boundary formula to the first-stage treated and untreated regions. The first-stage boundary terms cancel because their combined weight is $A_1(s) - A_0(s) = \kappa(s)$, which is zero on M_1. Hence the total welfare has no first-order M_1 boundary contribution.

Substituting the formulas for $\dot{A}_1$ and $\dot{A}_0$ gives

$$\begin{aligned}
\dot{V}^* &= \int_{D_1^+} \int_{D_2^+} \dot{\mu}_1(x)p_1(x \mid s)m(s)\,\mathrm{d}x\,\mathrm{d}s + \int_{D_1^+} \int_{D_2^-} \dot{\mu}_0(x)p_1(x \mid s)m(s)\,\mathrm{d}x\,\mathrm{d}s \\
&\quad + \int_{D_1^-} \int_{D_2^+} \dot{\mu}_1(x)p_0(x \mid s)m(s)\,\mathrm{d}x\,\mathrm{d}s + \int_{D_1^-} \int_{D_2^-} \dot{\mu}_0(x)p_0(x \mid s)m(s)\,\mathrm{d}x\,\mathrm{d}s.
\end{aligned}$$

The derivative of the total welfare contains only full-dimensional Lebesgue integral terms.

Thus, with respect to the perturbations in μ_0 and μ_1 considered here, the total welfare has an ordinary L_2-type first-order derivative, whereas the individual path components generally contain lower-dimensional boundary functionals.

The same cancellation can be seen directly by differentiating the four path components and summing them. Define

$$G_{11}(s) := \int_{D_2^+} \mu_1(x)p_1(x \mid s)\,\mathrm{d}x, \qquad G_{10}(s) := \int_{D_2^-} \mu_0(x)p_1(x \mid s)\,\mathrm{d}x,$$
$$G_{01}(s) := \int_{D_2^+} \mu_1(x)p_0(x \mid s)\,\mathrm{d}x, \qquad G_{00}(s) := \int_{D_2^-} \mu_0(x)p_0(x \mid s)\,\mathrm{d}x.$$

Then

$$A_1(s) = G_{11}(s) + G_{10}(s), \qquad A_0(s) = G_{01}(s) + G_{00}(s),$$

and

$$V_{11}^* = \int_{D_1^+} G_{11}(s)m(s)\,\mathrm{d}s, \qquad V_{10}^* = \int_{D_1^+} G_{10}(s)m(s)\,\mathrm{d}s,$$
$$V_{01}^* = \int_{D_1^-} G_{01}(s)m(s)\,\mathrm{d}s, \qquad V_{00}^* = \int_{D_1^-} G_{00}(s)m(s)\,\mathrm{d}s.$$

The derivatives of the second-stage component functions are

$$\dot{G}_{11}(s) = \int_{D_2^+} \dot{\mu}_1(x)p_1(x \mid s)\,\mathrm{d}x + \int_{M_2} \frac{\dot{\delta}(x)\mu_1(x)p_1(x \mid s)}{\|\nabla_x \delta(x)\|}\,\mathrm{d}\mathcal{H}^{d_x-1}(x),$$
$$\dot{G}_{10}(s) = \int_{D_2^-} \dot{\mu}_0(x)p_1(x \mid s)\,\mathrm{d}x - \int_{M_2} \frac{\dot{\delta}(x)\mu_0(x)p_1(x \mid s)}{\|\nabla_x \delta(x)\|}\,\mathrm{d}\mathcal{H}^{d_x-1}(x),$$
$$\dot{G}_{01}(s) = \int_{D_2^+} \dot{\mu}_1(x)p_0(x \mid s)\,\mathrm{d}x + \int_{M_2} \frac{\dot{\delta}(x)\mu_1(x)p_0(x \mid s)}{\|\nabla_x \delta(x)\|}\,\mathrm{d}\mathcal{H}^{d_x-1}(x),$$
$$\dot{G}_{00}(s) = \int_{D_2^-} \dot{\mu}_0(x)p_0(x \mid s)\,\mathrm{d}x - \int_{M_2} \frac{\dot{\delta}(x)\mu_0(x)p_0(x \mid s)}{\|\nabla_x \delta(x)\|}\,\mathrm{d}\mathcal{H}^{d_x-1}(x).$$

The plus signs correspond to derivatives of treated second-stage regions $D_{2,t}^+$, and the minus signs correspond to derivatives of untreated second-stage regions $D_{2,t}^-$. These signs are purely geometric: when $D_{2,t}^+$ expands, $D_{2,t}^-$ contracts.

Applying the first-stage moving-boundary formula gives

$$
\begin{aligned}
\dot{V}_{11}^{*} &= \int_{D_1^+} \dot{G}_{11}(s)m(s)\,\mathrm{d}s + \int_{M_1} \frac{\dot{\kappa}(s)G_{11}(s)m(s)}{\|\nabla_s\kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s) \\
&= \int_{D_1^+}\int_{D_2^+} \dot{\mu}_1(x)p_1(x \mid s)m(s)\,\mathrm{d}x\,\mathrm{d}s \\
&\quad + \int_{D_1^+}\int_{M_2} \frac{\dot{\delta}(x)\mu_1(x)p_1(x \mid s)}{\|\nabla_x\delta(x)\|}m(s)\,\mathrm{d}\mathcal{H}^{d_x-1}(x)\,\mathrm{d}s \\
&\quad + \int_{M_1} \frac{\dot{\kappa}(s)G_{11}(s)m(s)}{\|\nabla_s\kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s), \\
\dot{V}_{10}^{*} &= \int_{D_1^+} \dot{G}_{10}(s)m(s)\,\mathrm{d}s + \int_{M_1} \frac{\dot{\kappa}(s)G_{10}(s)m(s)}{\|\nabla_s\kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s) \\
&= \int_{D_1^+}\int_{D_2^-} \dot{\mu}_0(x)p_1(x \mid s)m(s)\,\mathrm{d}x\,\mathrm{d}s \\
&\quad - \int_{D_1^+}\int_{M_2} \frac{\dot{\delta}(x)\mu_0(x)p_1(x \mid s)}{\|\nabla_x\delta(x)\|}m(s)\,\mathrm{d}\mathcal{H}^{d_x-1}(x)\,\mathrm{d}s \\
&\quad + \int_{M_1} \frac{\dot{\kappa}(s)G_{10}(s)m(s)}{\|\nabla_s\kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s), \\
\dot{V}_{01}^{*} &= \int_{D_1^-} \dot{G}_{01}(s)m(s)\,\mathrm{d}s - \int_{M_1} \frac{\dot{\kappa}(s)G_{01}(s)m(s)}{\|\nabla_s\kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s) \\
&= \int_{D_1^-}\int_{D_2^+} \dot{\mu}_1(x)p_0(x \mid s)m(s)\,\mathrm{d}x\,\mathrm{d}s \\
&\quad + \int_{D_1^-}\int_{M_2} \frac{\dot{\delta}(x)\mu_1(x)p_0(x \mid s)}{\|\nabla_x\delta(x)\|}m(s)\,\mathrm{d}\mathcal{H}^{d_x-1}(x)\,\mathrm{d}s \\
&\quad - \int_{M_1} \frac{\dot{\kappa}(s)G_{01}(s)m(s)}{\|\nabla_s\kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s), \\
\dot{V}_{00}^{*} &= \int_{D_1^-} \dot{G}_{00}(s)m(s)\,\mathrm{d}s - \int_{M_1} \frac{\dot{\kappa}(s)G_{00}(s)m(s)}{\|\nabla_s\kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s) \\
&= \int_{D_1^-}\int_{D_2^-} \dot{\mu}_0(x)p_0(x \mid s)m(s)\,\mathrm{d}x\,\mathrm{d}s \\
&\quad - \int_{D_1^-}\int_{M_2} \frac{\dot{\delta}(x)\mu_0(x)p_0(x \mid s)}{\|\nabla_x\delta(x)\|}m(s)\,\mathrm{d}\mathcal{H}^{d_x-1}(x)\,\mathrm{d}s \\
&\quad - \int_{M_1} \frac{\dot{\kappa}(s)G_{00}(s)m(s)}{\|\nabla_s\kappa(s)\|}\,\mathrm{d}\mathcal{H}^{d_s-1}(s).
\end{aligned}
$$

The preceding displays use the same moving-boundary calculation twice: first for the second-stage regions, and then for the first-stage regions. No new identification argument is used; these are algebraic expansions of the pathwise derivatives of the four identified component functionals.

Let $B_{ab}^{M_2}$ denote the M_2 boundary term in $\dot{V}_{ab}^{*}$, and let $B_{ab}^{M_1}$ denote the M_1 boundary term. The second-stage boundary terms satisfy

$$
\begin{aligned}
&B_{11}^{M_2} + B_{10}^{M_2} + B_{01}^{M_2} + B_{00}^{M_2} \\
&= \int_{D_1^+} \int_{M_2} \frac{\dot{\delta}(x)\mu_1(x)p_1(x \mid s)}{\|\nabla_x \delta(x)\|} m(s) \, \mathrm{d}\mathcal{H}^{d_x - 1}(x) \, \mathrm{d}s \\
&\quad - \int_{D_1^+} \int_{M_2} \frac{\dot{\delta}(x)\mu_0(x)p_1(x \mid s)}{\|\nabla_x \delta(x)\|} m(s) \, \mathrm{d}\mathcal{H}^{d_x - 1}(x) \, \mathrm{d}s \\
&\quad + \int_{D_1^-} \int_{M_2} \frac{\dot{\delta}(x)\mu_1(x)p_0(x \mid s)}{\|\nabla_x \delta(x)\|} m(s) \, \mathrm{d}\mathcal{H}^{d_x - 1}(x) \, \mathrm{d}s \\
&\quad - \int_{D_1^-} \int_{M_2} \frac{\dot{\delta}(x)\mu_0(x)p_0(x \mid s)}{\|\nabla_x \delta(x)\|} m(s) \, \mathrm{d}\mathcal{H}^{d_x - 1}(x) \, \mathrm{d}s \\
&= \int_{D_1^+} \int_{M_2} \frac{\dot{\delta}(x)\{\mu_1(x) - \mu_0(x)\}p_1(x \mid s)}{\|\nabla_x \delta(x)\|} m(s) \, \mathrm{d}\mathcal{H}^{d_x - 1}(x) \, \mathrm{d}s \\
&\quad + \int_{D_1^-} \int_{M_2} \frac{\dot{\delta}(x)\{\mu_1(x) - \mu_0(x)\}p_0(x \mid s)}{\|\nabla_x \delta(x)\|} m(s) \, \mathrm{d}\mathcal{H}^{d_x - 1}(x) \, \mathrm{d}s \\
&= \int_{D_1^+} \int_{M_2} \frac{\dot{\delta}(x)\delta(x)p_1(x \mid s)}{\|\nabla_x \delta(x)\|} m(s) \, \mathrm{d}\mathcal{H}^{d_x - 1}(x) \, \mathrm{d}s \\
&\quad + \int_{D_1^-} \int_{M_2} \frac{\dot{\delta}(x)\delta(x)p_0(x \mid s)}{\|\nabla_x \delta(x)\|} m(s) \, \mathrm{d}\mathcal{H}^{d_x - 1}(x) \, \mathrm{d}s \\
&= 0.
\end{aligned}
$$

The cancellation follows because $\delta(x) = 0$ on M_2. Thus the second-stage boundary terms are present in the individual components but vanish in their sum.

Similarly, the first-stage boundary terms satisfy

$$
\begin{aligned}
&B_{11}^{M_1} + B_{10}^{M_1} + B_{01}^{M_1} + B_{00}^{M_1} \\
&= \int_{M_1} \frac{\dot{\kappa}(s) G_{11}(s) m(s)}{\|\nabla_s \kappa(s)\|} \, \mathrm{d}\mathcal{H}^{d_s - 1}(s) + \int_{M_1} \frac{\dot{\kappa}(s) G_{10}(s) m(s)}{\|\nabla_s \kappa(s)\|} \, \mathrm{d}\mathcal{H}^{d_s - 1}(s) \\
&\quad - \int_{M_1} \frac{\dot{\kappa}(s) G_{01}(s) m(s)}{\|\nabla_s \kappa(s)\|} \, \mathrm{d}\mathcal{H}^{d_s - 1}(s) - \int_{M_1} \frac{\dot{\kappa}(s) G_{00}(s) m(s)}{\|\nabla_s \kappa(s)\|} \, \mathrm{d}\mathcal{H}^{d_s - 1}(s) \\
&= \int_{M_1} \frac{\dot{\kappa}(s) \{G_{11}(s) + G_{10}(s)\} m(s)}{\|\nabla_s \kappa(s)\|} \, \mathrm{d}\mathcal{H}^{d_s - 1}(s) \\
&\quad - \int_{M_1} \frac{\dot{\kappa}(s) \{G_{01}(s) + G_{00}(s)\} m(s)}{\|\nabla_s \kappa(s)\|} \, \mathrm{d}\mathcal{H}^{d_s - 1}(s) \\
&= \int_{M_1} \frac{\dot{\kappa}(s) \{A_1(s) - A_0(s)\} m(s)}{\|\nabla_s \kappa(s)\|} \, \mathrm{d}\mathcal{H}^{d_s - 1}(s) \\
&= \int_{M_1} \frac{\dot{\kappa}(s) \kappa(s) m(s)}{\|\nabla_s \kappa(s)\|} \, \mathrm{d}\mathcal{H}^{d_s - 1}(s) \\
&= 0.
\end{aligned}
$$

The cancellation follows because $\kappa(s) = 0$ on M_1. This is the first-stage analogue of the cancellation of the M_2 terms.

After these cancellations, the derivative of the sum of the four components is

$$
\begin{aligned}
\dot{V}^* &= \dot{V}_{11}^* + \dot{V}_{10}^* + \dot{V}_{01}^* + \dot{V}_{00}^* \\
&= \int_{D_1^+} \int_{D_2^+} \dot{\mu}_1(x) p_1(x \mid s) m(s) \, \mathrm{d}x \, \mathrm{d}s + \int_{D_1^+} \int_{D_2^-} \dot{\mu}_0(x) p_1(x \mid s) m(s) \, \mathrm{d}x \, \mathrm{d}s \\
&\quad + \int_{D_1^-} \int_{D_2^+} \dot{\mu}_1(x) p_0(x \mid s) m(s) \, \mathrm{d}x \, \mathrm{d}s + \int_{D_1^-} \int_{D_2^-} \dot{\mu}_0(x) p_0(x \mid s) m(s) \, \mathrm{d}x \, \mathrm{d}s.
\end{aligned}
$$

This agrees with the direct derivative of the total welfare derived above. The component-by-component calculation is useful because it shows exactly where the apparent irregularity goes: it is not absent from the component derivatives, but it cancels after aggregation.

This calculation also clarifies the distinction between welfare and value functionals. For individual path components, the moving-boundary terms have nonzero weights such as $\mu_1(x)$, $\mu_0(x)$, or $G_{ab}(s)$ on the relevant margins. These are lower-dimensional Hausdorff integral functionals and are generally irregular in the sense of Chen and Gao (2026). For the total welfare, the second-stage boundary weight reduces to $\mu_1(x) - \mu_0(x) = \delta(x)$ on M_2, and the first-stage boundary weight reduces to $A_1(s) - A_0(s) = \kappa(s)$ on M_1. Both weights vanish on their respective margins. This is the dynamic analogue of the distinction between regular welfare functionals and irregular value functionals emphasized in the optimal treatment assignment literature.

The preceding derivative calculations are valid only under thin, regular margins. In particular, $M_2 = \{x : \delta(x) = 0\}$ must have zero d_x-dimensional Lebesgue measure and must satisfy $\|\nabla_x \delta(x)\| \neq 0$ on the margin. Similarly, $M_1 = \{s : \kappa(s) = 0\}$ must have zero d_s-dimensional Lebesgue measure and must satisfy $\|\nabla_s \kappa(s)\| \neq 0$ on the margin. Smoothness of M_1 should be imposed through smoothness and regularity of $s \mapsto \kappa(s)$; it is not automatic merely because V_2 is integrated over X. Sufficient smoothness of $p_a(x \mid s)$, $\mu_a(x)$, and the support conditions are needed for this step.

If the regular-margin conditions fail, the moving-boundary expressions with denominators $\|\nabla_x \delta(x)\|$ and $\|\nabla_s \kappa(s)\|$ need not be valid. For example, the expansion can fail if $\{\delta(x) = 0\}$ has positive d_x-dimensional Lebesgue measure, if $\nabla_x \delta(x) = 0$ on the second-stage margin, if $\{\kappa(s) = 0\}$ has positive d_s-dimensional Lebesgue measure, or if $\nabla_s \kappa(s) = 0$ on the first-stage margin. In such cases, the relevant path components may be only directionally differentiable, or may fail to have a linear pathwise derivative. Even when the derivative exists, a component derivative containing Hausdorff integrals over M_1 or M_2 is generally not an ordinary L_2-regular derivative, because it depends on the perturbation through lower-dimensional sets. This is the thin-set irregularity emphasized by Chen and Gao (2026). By contrast, the total welfare derivative above has no such lower-dimensional term after the cancellations, which explains why the total welfare can be regular even though its individual treatment-path components are not.

2.5 Generalized Dynamic Treatment Regime (Sakaguchi, 2025)

3 Literature Review

My paper is conceptually similar to Viviano and Bradic, where they study a two step approach, finding pareto prontier and then minimize unfairness. They proceed in a lexicographic manner. Viviano and Bradic (2024)

4 Model Setup and Directional Derivatives

4.1 Social Planner's Problem

$$\begin{aligned} \min_{\pi(\cdot)} \quad & \mathbb{E}\left[Y(1)\pi(X) + Y(0)(1-\pi(X))\right] \\ \text{s.t.} \quad & \mathbb{E}[Z(1)\pi(X) + Z(0)(1-\pi(X))] \geq 0 \end{aligned}$$

This is equivalent to

$$\begin{aligned} \min_{\pi(\cdot)} \quad & \mathbb{E}\left[Y(1)\pi(X) + Y(0)(1-\pi(X)) - \lambda(Z(1)\pi(X) + Z(0)(1-\pi(X)))\right] \\ & \mathbb{E}[(Y(1) - \lambda Z(1))\pi(X) + (Y(0) - \lambda Z(0))(1-\pi(X))] \end{aligned}$$

so the optimal rule is

$$\mathbb{1}\{\mathbb{E}[Y(1) - Y(0) - \lambda(Z(1) - Z(0)) \mid X] \geq 0\}$$

what is λ? it quantifies trade off
If λ is 0, then

4.2 Social Planner's Problem

Consider a social planner's idealized problem with a two-item checklist: In the first stage

$$\min_{p(\cdot)} \mathbb{E}\left[Z_i(1)p(X) + Z_i(0)(1-p(X))\right]$$

which induces a hard-threshold constraint $\mathbb{1}\{g_0(x) \geq 0\}$

In the second stage

$$\mathbb{E}[Y_i(1)\mathbb{1}\{h_0(X) \geq 0, g_0(X) \geq 0\}]$$

5 Estimation and Inference of the Value Functional

References

Chen, Q., Austern, M. and Syrgkanis, V. (2023). Inference on optimal dynamic policies via softmax approximation. *arXiv preprint arXiv:2303.04416*.

Chen, X. and Gao, W. Y. (2026). Thin Sets Are Not Equally Thin: Minimax Learning of Submanifold Integrals. Working paper. First version: arXiv:2507.12673v1.

Sakaguchi, S. (2025). Estimation of optimal dynamic treatment assignment rules under policy constraints. *Quantitative Economics*, **16** (3), 981–1022.

Viviano, D. and Bradic, J. (2024). Fair policy targeting. *Journal of the American Statistical Association*, **119** (545), 730–743.

A Proofs

A.1 Proof of Proposition 2.2

$$\max_{\pi} \mathbb{E}\left[Y^{(\pi)}\right] = \max_{\pi} \mathbb{E}\left[\mathbb{E}\left[Y^{(\pi)} \ \mid \ S\right]\right] = \max_{\pi_2} \mathbb{E}\left[\max_{\tau_1} \mathbb{E}\left[Y^{(\tau_1,\pi_2)} \mid S\right]\right]$$

where the second equation follows because the first period τ_1 is not allowed to depend on the second period policy π_2. Thus the optimal first-period policy is defined as

$$\pi_1^*(S) = \arg\max_{\tau_1} \ \mathbb{E}\left[Y^{(\tau_1,\pi_2^*)} \mid S\right]$$

Moreover, we can further write

$$\begin{aligned}
\max_{\pi_2} \mathbb{E}\left[\max_{\tau_1} \mathbb{E}\left[Y^{(\tau_1,\pi_2)} \mid S\right]\right] &= \max_{\pi_2} \mathbb{E}\left[\max_{\tau_1} \mathbb{E}\left[Y^{(\tau_1,\pi_2)} \mid T_1 = \tau_1, S\right]\right] \\
&= \max_{\pi_2} \mathbb{E}\left[\max_{\tau_1} \mathbb{E}\left[Y^{(T_1,\pi_2)} \mid T_1 = \tau_1, S\right]\right] \\
&= \max_{\pi_2} \mathbb{E}\left[\max_{\tau_1} \mathbb{E}\left[\mathbb{E}\left[Y^{(T_1,\pi_2)} \mid X, T_1 = \tau_1, S\right] \mid T_1 = \tau_1, S\right]\right] \\
&= \max_{\pi_2} \mathbb{E}\left[\max_{\tau_1} \mathbb{E}\left[\mathbb{E}\left[Y^{(T_1,\pi_2)} \mid X\right] \mid T_1 = \tau_1, S\right]\right] \\
&= \mathbb{E}\left[\max_{\tau_1} \mathbb{E}\left[\max_{\tau_2} \mathbb{E}\left[Y^{(T_1,\tau_2)} \mid X\right] \mid T_1 = \tau_1, S\right]\right]
\end{aligned}$$

Thus, the optimal second period policy is defined as

$$\pi_2^*(X) = \arg\max_{\tau_2} \ \mathbb{E}\left[Y^{(T_1,\tau_2)} \mid X\right]$$

This implies

$$\begin{aligned}
\mathbb{1}\{\pi_2^*(X) = 1\} &= \mathbb{1}\left\{\mathbb{E}\left[Y^{(T_1,1)} \mid X\right] - \mathbb{E}\left[Y^{(T_1,0)} \mid X\right] \geq 0\right\} \\
&= \mathbb{1}\left\{\mathbb{E}\left[Y^{(T_1,1)} \mid X, T_2 = 1\right] - \mathbb{E}\left[Y^{(T_1,0)} \mid X, T_2 = 0\right] \geq 0\right\} \\
&= \mathbb{1}\left\{\mathbb{E}\left[Y \mid X, T_2 = 1\right] - \mathbb{E}\left[Y \mid X, T_2 = 0\right] \geq 0\right\} \\
&= \mathbb{1}\left\{\mu_1(X) - \mu_0(X)\right\} \geq 0
\end{aligned}$$

and

$$
\begin{aligned}
\mathbb{1}\{\pi_1^*(S)=1\} &= \mathbb{1}\left\{\mathbb{E}\left[Y^{(1,\pi_2^*)} \mid S\right] - \mathbb{E}\left[Y^{(0,\pi_2^*)} \mid S\right] \geq 0\right\} \\
&= \mathbb{1}\left\{\mathbb{E}\left[Y^{(1,\pi_2^*)} \mid S, T_1=1\right] - \mathbb{E}\left[Y^{(0,\pi_2^*)} \mid S, T_1=0\right] \geq 0\right\} \\
&= \mathbb{1}\left\{\mathbb{E}\left[\mathbb{E}\left[Y^{(1,\pi_2^*)} \mid S, T_1=1, X\right] \mid S, T_1=1\right] - \mathbb{E}\left[\mathbb{E}\left[Y^{(0,\pi_2^*)} \mid S, T_1=0, X\right] \mid S, T_1=0\right] \geq 0\right\} \\
&= \mathbb{1}\left\{\mathbb{E}\left[\mathbb{E}\left[Y^{(T_1,\pi_2^*)} \mid S, T_1=1, X\right] \mid S, T_1=1\right] - \mathbb{E}\left[\mathbb{E}\left[Y^{(T_1,\pi_2^*)} \mid S, T_1=0, X\right] \mid S, T_1=0\right] \geq
\\
&= \mathbb{1}\left\{\mathbb{E}\left[\mathbb{E}\left[Y^{(T_1,\pi_2^*)} \mid X\right] \mid S, T_1=1\right] - \mathbb{E}\left[\mathbb{E}\left[Y^{(T_1,\pi_2^*)} \mid X\right] \mid S, T_1=0\right] \geq 0\right\} \\
&= \mathbb{1}\left\{\mathbb{E}\left[\max_{\tau_2} \mathbb{E}\left[Y^{(T_1,\tau_2)} \mid X\right] \mid S, T_1=1\right] - \mathbb{E}\left[\max_{\tau_2} \mathbb{E}\left[Y^{(T_1,\tau_2)} \mid X\right] \mid S, T_1=0\right] \geq 0\right\} \\
&= \mathbb{1}\left\{\mathbb{E}[V_2(X) \mid S, T_1=1] - \mathbb{E}[V_2(X) \mid S, T_1=0] \geq 0\right\}
\end{aligned}
$$